

Siddharth Behera

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PROFILE

Results-driven Software Engineer with hands-on experience building scalable, high-performance web applications using the MERN stack. Passionate about creating intuitive user experiences with React (Redux, React Router) and developing reliable backend systems using Node.js and Express.js.

Currently expanding expertise in System Design and modern software architecture, including Microservices, Docker, Kubernetes, and event-driven systems. Driven by solving complex problems, optimizing application performance, and building scalable, maintainable software solutions.

PROFESSIONAL EXPERIENCE

Freelance Full Stack Developer (Contract) | Self-Employed

- Developed and maintained full-stack features using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Designed and implemented RESTful APIs using Node.js and Express.js.
- Optimized MongoDB queries through indexing and aggregation pipeline improvements.
- Integrated Redis caching to improve application performance and response times.
- Implemented secure authentication and authorization using JWT and bcrypt.js.
- Containerized applications using Docker and worked with Kubernetes for deployment and orchestration.
- Built and maintained CI/CD pipelines using Jenkins to automate build and deployment workflows.
- Collaborated with stakeholders to understand requirements, implement features, and resolve production issues.

PROJECT

Chat Application – “Talk-A-Tive”

Deployment: Render

Link: <https://talk-a-tive-pwvu.onrender.com/>

Description:

Talk-A-Tive is a real-time chat web application built using the MERN stack with secure user authentication. It allows users to chat one-on-one or in groups, create channels, and manage participants. The application leverages WebSockets via Socket.io for seamless, real-time messaging and provides a responsive interface for both desktop and mobile users.

Summary:

Developed a full-featured real-time chat application with secure user authentication, personal and group chat management, and user search capabilities. The app supports adding or removing members, renaming group chats, and includes real-time notifications with typing indicators. Designed with a responsive and fluid user interface optimized for both desktop and mobile devices.

EDUCATION

Central Tool Room & Training Centre (CTTC), Bhubaneswar – Post-Graduate Diploma, Tool Design and CAD/CAM

Central Institute of Plastics Engineering & Technology (CIPET), Bhubaneswar – B.Tech, Plastic Engineering

TECH STACK

Frontend	HTML5, CSS3, JavaScript, TypeScript
Libraries & UI	React (Redux, React Router), Bootstrap, Chakra UI, Hope UI, Axios, Socket.io-client
Backend	Node.JS, Express.JS, JWT, Bcrypt.js, Socket.io
Database	MongoDB, Redis
ORM/ODM	Mongoose
DevOps / Microservices	Docker, Kubernetes, Kafka
CI/CD	GitHub Actions
Tools & Utilities	Git, GitHub, Postman, Firebase
IDE / Editor	Visual Studio Code

SOFTWARE ARCHITECTURE & DESIGN CONCEPTS

As an emerging professional, I have developed a good understanding of critical concepts and tools, including...

- **Networking Basics:** Understanding of HTTP/HTTPS protocols, REST APIs, WebSockets, DNS resolution, and TCP/UDP communication
- **Software Architecture:** Understanding Microservices, and Monolithic vs. Distributed architectural patterns
- **Nginx:** Basic configuration, reverse proxy, load balancing
- **Containerization & Orchestration:** Managing Docker containers and orchestrating Kubernetes clusters and pods for scalable deployments
- **Microservices architecture:** Understanding service decomposition, inter-service communication, and scalability benefits
- **DevOps Principles:** Implementation of Continuous Integration (CI), Continuous Deployment (CD), and Infrastructure as Code (IaC) practices for streamlined development and delivery
- **System design Fundamentals:** Designing for scalability (vertical and horizontal scaling), fault tolerance, high availability, reliability, redundancy, and optimal throughput
- **Messaging systems:** Understanding of Message Queues (e.g., Kafka) and event-driven architectures for asynchronous communication
- **Security Fundamentals:** Authentication, Authorization, JWT, Encryption
- **Database Concepts:** Strong understanding of SQL and NoSQL databases, ACID transactions, data modeling, indexing strategies, caching with Redis, as well as replication and sharding for scalability and reliability
- **Advanced Architectural Concepts:** Understanding of CAP and PACELC theorems, consistent hashing for distributed systems, proxies and load balancers for traffic management, and CDN for content delivery optimization
- **API Design & Communication Paradigms:** Designing and integrating APIs using REST, GraphQL, and gRPC, with a focus on backward compatibility and maintainability
- **Monitoring & Observability:** Implementing logging, metrics, and alerting systems, with integration into external monitoring services for performance tracking and issue resolution
- **Object Storage & External Services:** Designing and integrating scalable object storage systems and cloud-based external APIs for data persistence and interoperability

PROJECT

Connect-X – Real-Time Communication Platform

Deployment: Render

Link: <https://connet-x674.onrender.com>

Description:

Connet-X is an ongoing personal project built with a strong emphasis on system design, scalable microservices, and real-time communication. The project was rapidly prototyped using GitHub Copilot AI, allowing more focus on architecture, distributed communication patterns, and performance considerations.

Key Features:

- Google OAuth for secure authentication
- Peer-to-peer (P2P) file transfer using WebRTC
- High-quality real-time video calling
- Live user presence and messaging powered by Socket.io
- Profile picture upload and update
- Separated services to reflect microservices patterns (auth, communication, file handling, etc.)

Summary:

Designed as a learning-focused yet production-like application to deepen expertise in system design, WebRTC, and event-driven real-time communication workflows. The platform showcases scalable architecture choices, efficient state handling, and seamless real-time interactions.

PERSONAL DETAILS

Location: Nagpur, India

Portfolio: <https://devsiddharthbehera.in>

Tech Interests: Distributed Systems, System Design

Soft Skills: Problem-solving, Collaboration, Ownership, Fast Learning